

What should K be?

A first simulation of priority applications in an AI-era job market

Daniel Bulla, Bratislava · September 2026 · Working note. A toy model with illustrative parameters, not calibrated to any real market. Its purpose is to show which numbers decide K. The simulation was built and run with an AI assistant (Claude, by Anthropic) under my direction.

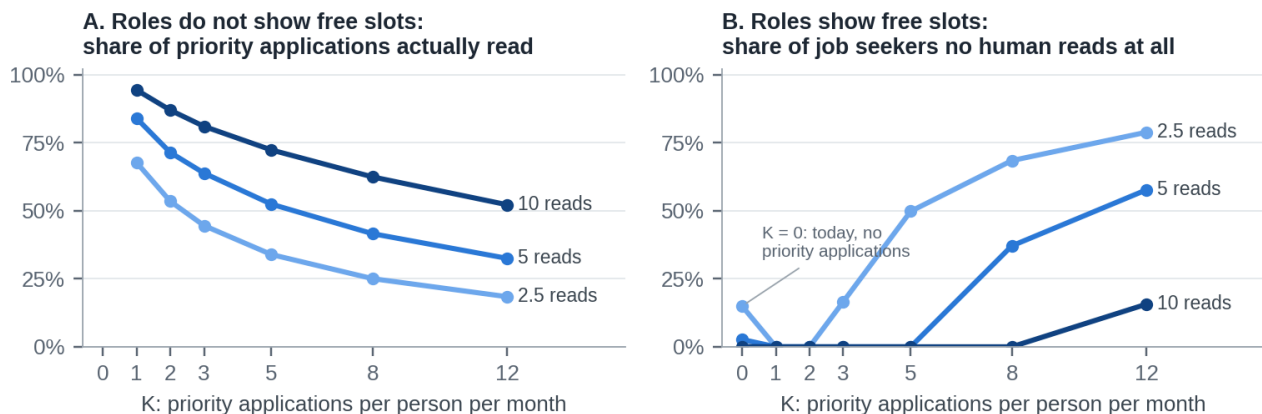
The question

In the General Matchmaking Protocol each person gets a small number, K, of **priority applications** a month. Each role promises that a named person will read every priority application and reply by a set date. Ordinary applications still exist. K = 5 was a placeholder. If K is too low, people cannot reach enough roles. If it is too high, the signal becomes cheap again and roles cannot keep their promise. One real market already runs K = 1: since June 2025 Greenhouse has let job seekers flag one "dream job" a month across its platform, and it reports those applicants are hired at roughly five times the usual rate [4]. That is a correlation, not a test of the right K, but it shows the mechanism works at scale.

The model

- 3,000 job seekers and 300 roles, one opening each, over one month.
- Everyone sends 30 ordinary applications, chosen by their own assistant from how much they want each role (some roles are popular with everyone) and a private, noisy check of their fit.
- Each role has **H careful human reads** a month and interviews 5 people. Without priority it chooses what to read with a noisy automated screen.
- With priority, each person marks K applications as priority. A role reads all of those first, then uses any spare time on the screen's picks. Roles make offers in order of fit; people accept the offer they like best.
- Two versions: roles either hide or **show how many priority slots they have left**.

One number summarises the market: $c = H \times J / N$, the careful reads available per job seeker per month (J roles, N job seekers). The model was run with $c = 1, 2.5, 5$ and 10 , each with K from 0 to 12, six runs each.



Lines: careful human reads available per job seeker per month ($c = H \times J / N$). Mean of 6 runs, 3,000 job seekers, 300 roles.

Three findings

1. **The deciding number is c, and the best K sat between half of c and c itself.** With free slots shown, the K that gave people their most-preferred roles, while leaving nobody unread, was 1 at $c = 1$, 2 at $c = 2.5$, 3 at $c = 5$ and 5 at $c = 10$. Beyond that, priority applications use up all reading time, the ordinary channel stops being read, and people spend priority on roles far down their own list. At $c = 5$ the average priority application went to role number 4 on the person's list at $K = 3$, and to number 21 at $K = 5$.
2. **K alone cannot keep the promise. Showing free slots can.** When roles hide how full they are, demand piles onto popular roles and promises break even at $K = 1$ (panel A). At $c = 10$ and $K = 5$, the share of broken promises ranged from 4% to 54% depending only on how concentrated demand was. Showing free slots keeps every promise by design. The cost appears once K passes c: then some people are never read at all (panel B).
3. **Job seekers gain; employers gain a little and give up a little.** At $c = 5$ and $K = 3$, people ended up in roles they ranked higher (position 7.9 on their list against 10.2 today), nobody went unread, and employers needed fewer offers per hire (1.32 against 1.49). The cost: employers' average fit with their hires fell 3%, because some reading moved from the screen's top picks to priority applicants. This matches the direction of a theory result on one signal per worker: worker welfare and matches rise, the effect on firms is ambiguous [1].

Detail at $c = 5$ (50 careful reads per role, 10 job seekers per role)

K	Promises kept if roles hide free slots	Job seekers never read (slots shown)	Where people end up: position of the hired role on their own list	Offers per hire	Employer's fit with hires (K = 0 is 100)
0 (today)	n/a	3%	10.2	1.49	100
1	84%	0%	9.4	1.43	100
3	64%	0%	7.9	1.32	97
5	52%	0%	12.1	1.25	94
8	42%	37%	13.0	1.46	89

$K = 5$ already overshoots here: promises hold with slots shown, but people are pushed down their own lists. Fit is in model units, indexed to today.

What this means for K

K should be a market parameter, not a constant. $K = 5$ fits a market that can give each job seeker about ten careful reads a month. In tighter markets it should be 2 or 3. A rule worth testing: set K near $0.6 \times c$ for each market (at least 1), and have every role show its free priority slots. The slots do more work than the exact value of K .

What the model leaves out

- It covers one static month. There is no learning across months, no holding priority back, no extra accounts, and no employer overstating its reading time.
- It overstates how much priority raises interview chances, because priority guarantees a read. In the economists' job market, applications with a signal led to an interview 31.5% of the time, against 24.7% without [2].
- Free slots go first come, first served. When they run out, late arrivals lose. Weekly batches in random order would be fairer; that is untested.
- It does not count the reading time employers save by receiving fewer blind applications, which is the main benefit claimed for them. Changing the noise of the automated screen hardly changed the results; changing the concentration of demand changed them a lot.

The data that would calibrate it

Three numbers, for any real market: (1) how many applications a job seeker sends a month and what share gets any reply; (2) how many applications a role can read carefully; (3) how unevenly applications spread across roles. Averages are not enough. Profesia.sk reports 29 responses per ad in 2025 [3], but the spread across ads is what breaks promises. If you hold data like this, or know who does, I would like to hear from you.

Python code (about 150 lines) and full results are available on request. Contact: dbullak@gmail.com

Sources

[1] Coles, Kushnir and Niederle (2013), "Preference Signaling in Matching Markets", American Economic Journal: Microeconomics 5(2), 99 to 134.

[2] Coles, Cawley, Levine, Niederle, Roth and Siegfried (2010), "The Job Market for New Economists: A Market Design Perspective", Journal of Economic Perspectives 24(4), Table 3.

[3] Profesia.sk annual statistics for 2025, as reported by oPeniazoch.sk (2026).

[4] Greenhouse, "Greenhouse Launches Dream Job" (5 June 2025); Fortune, 27 July 2026, reporting Greenhouse's figures.